Companion to **Cube-root growth of most non-highway generalised Langton's ants of length up to 12**.

Audit numbers match the references in the main paper.

Supplementary material: Reproducibility & Preregistration Audit Table

This audit preserves the repository locations displaced from the main narrative. All paths are relative to the repository root. Audit numbers link the physical tests to their records; they are not additional scientific results.

Legacy audit baseline (rows 1–43): 71ffb99863465b7dbc8f7912571af9c32626f339 (2026-10-15T22:41:51+13:00), including then-uncommitted manuscript and research edits. The October follow-up rows 44–48 pin evidence at revision 0de4d38. In legacy rows the Git field gives the earliest commit located for that path at audit time (hash and committer timestamp), not proof that the hypothesis was committed before measurement. Document dates are author-recorded dates; migration commits can be later. Missing seeds or registration evidence must not be inferred from these timestamps.

The source and recipe locations below retain commands, full seed schedules, windows, input hashes and exclusions where recorded. Seed excerpts are quoted literally from those sources; they may describe sampling or bootstrap seeds rather than walker seeds. Blank-board deterministic runs need no random seed. "Not located" means this audit did not recover an explicit seed declaration, not that a run was unseeded.

Audit	Exact records and producers	Seeds, registration and Git record
1	experiments/2026-10-13-census-sweeps/results/sweep_v2.csv experiments/2026-10-13-census-sweeps/scripts/census.py experiments/2026-10-13-census-sweeps/results/unresolved_le10.txt experiments/2026-10-13-census-sweeps/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-census-sweeps/scripts/census.py experiments/2026-10-13-census-sweeps/scripts/equiv_census.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
2	experiments/2026-10-13-probe-growth-census/results/growth.csv experiments/2026-10-13-probe-growth-census/results/probe_summary.csv experiments/2026-10-13-probe-growth-census/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-probe-growth-census/scripts/c_letters.py experiments/2026-10-13-probe-growth-census/scripts/c_model.py experiments/2026-10-13-probe-growth-census/scripts/cube_fill.py experiments/2026-10-13-probe-growth-census/scripts/cube_stats.py experiments/2026-10-13-probe-growth-census/scripts/sq_baseline_window.py	Seed declarations: rng = np.random.default_rng(20261012); idx = rng.permutation(len(c)); folds = np.array_split(idx, 5) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
3	experiments/2026-10-13-b4-unfitted-rules/results/b4_builders.txt experiments/2026-10-13-b4-unfitted-rules/results/b4_stats.txt experiments/2026-10-13-b4-unfitted-rules/README.md Analysis scripts (commands in recipe):	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-

Audit	Exact records and producers	Seeds, registration and Git record
	experiments/2026-10-13-b4-unfitted-rules/scripts/b4_builders.py experiments/2026-10-13-b4-unfitted-rules/scripts/b4_stats.py	frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
4	tests/ant.rs Analysis scripts (commands in recipe): tests/ant.rs	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
5	tests/probe.rs Analysis scripts (commands in recipe): tests/probe.rs	Seed declarations: let seed = 7; Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
6	research-outputs/cube-root-growth-paper/compute_numbers.py research-outputs/cube-root-growth-paper/numbers.json research-outputs/cube-root-growth-paper/README.md research-outputs/cube-root-growth-paper/figures.py Analysis scripts (commands in recipe): research-outputs/cube-root-growth-paper/compute_numbers.py research-outputs/cube-root-growth-paper/kappa_prediction_numbers.py research-outputs/cube-root-growth-paper/figures.py	Seed declarations: rng = np.random.default_rng(20261012) uv run research-outputs/cube-root-growth-paper/compute_numbers.py # data/* -> research-outputs/cube-root-growth-paper/numbers.json (bootstrap seed 20261012) in the script (reference ant, rotor with numpy seed 1) or read from `experiments/2026-10-13-census-sweeps/results/sweep_v2.csv` and `experiments/2026-10-13-census-sweeps/results/census.csv`. Registration/context records (combined result notes are not independent preregistration evidence): Mixing test on a random sample of rules — Written 2026-10-12, before running. Reason: the earlier mixing sample notes/mixing-random-sample.md Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Prospective transport forecast — Registered 2026-10-13 23:35 NZDT, before computing forecasts. Base commit notes/transport-forecast-2026-10-13.md Earliest path commit: 9118bb51a3d74d01c99bcac1141a6d99674b4237 (2026-10-14T18:35:50+13:00)
7	experiments/2026-10-13-h1-lacunarity-decomposition/results/h1_lacunarity.txt experiments/2026-10-13-h1-lacunarity-decomposition/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-h1-lacunarity-decomposition/scripts/h1_decomp.py experiments/2026-10-13-h1-lacunarity-decomposition/scripts/h1_lacunarity.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
8	experiments/2026-10-13-h3-scaling-collapse/results/h3_collapse.txt experiments/2026-10-13-h3-scaling-collapse/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-h3-scaling-collapse/scripts/h3_collapse.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md

Audit	Exact records and producers	Seeds, registration and Git record
		Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
9	notes/cube-root-hypothesis.md	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Why do chaotic ants grow as t^(1/3)? — Written 2026-10-10 07:45 NZDT, before any measurement beyond the census notes/cube-root-hypothesis.md Earliest path commit: 50206967c655891659ac 98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
10	experiments/2026-10-13-a1-kappa/results/a1b_icc.txt experiments/2026-10-13-a1-kappa/results/a1_kappa_ proxy.txt experiments/2026-10-13-a1-kappa/results/a1c_predict.txt experiments/2026-10-13-a1-kappa/results/a1_edge_q.txt experiments/2026-10-13-a1-kappa/results/a1d_stats.txt experiments/2026-10-13-a1-kappa/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-a1-kappa/scripts/a1_edge_q.py experiments/2026-10-13-a1-kappa/scripts/a1_kappa_ proxy.py experiments/2026-10-13-a1-kappa/scripts/a1b_icc.py experiments/2026-10-13-a1-kappa/scripts/a1c_predict.py experiments/2026-10-13-a1-kappa/scripts/a1d_stats.py	Seed declarations: Linear model on q = #R/n, n, letters 1..6 (R=1), run count, trained on a random half (seed 20261013) of the non-bursty blob rules of rng = np.random.default_rng(20261013); idx = rng.permutation(len(rules)); h = len(idx) // 2; tr, te = idx[:h], idx[h:] Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
11	experiments/2026-10-13-b2-between-rule-variance/results/ b2_probe.csv experiments/2026-10-13-b2-between-rule-variance/ README.md Analysis scripts (commands in recipe): experiments/2026-10-13-b2-between-rule-variance/scripts/ b2_anova.py experiments/2026-10-13-b2-between-rule-variance/scripts/ b2_sample.py experiments/2026-10-13-b2-between-rule-variance/scripts/ b2b_anova.py	Seed declarations: - `scripts/b2_sample.py`: B2: 100 random non-bursty blob rules (numpy seed 20261013), disjoint from the 40 rules of experiments/2026-10-13-patched-starts/ results/patched.csv, 12 patched """B2: 100 random non-bursty blob rules (numpy seed 20261013), disjoint from the 40 rules of experiments/2026-10-13-patched-starts/ results/patched.csv, 12 patched seeds each. rng = np.random.default_rng(20261013); pick = sorted(rng.choice(len(rules), 100, replace=False)) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
12	experiments/2026-10-kappa-rule-space/README.md experiments/2026-10-kappa-rule-space/results/w3c_ stats.txt experiments/2026-10-kappa-rule-space/results/w3c_ cross.txt Analysis scripts (commands in recipe): experiments/2026-10-kappa-rule-space/scripts/kcommon.py experiments/2026-10-kappa-rule-space/scripts/prefix_ audit.py experiments/2026-10-kappa-rule-space/scripts/prefix_audit_ notebook.py experiments/2026-10-kappa-rule-space/scripts/w0_build.py experiments/2026-10-kappa-rule-space/scripts/w1_ smooth.py experiments/2026-10-kappa-rule-space/scripts/w2_ stationarity.py experiments/2026-10-kappa-rule-space/scripts/w3b_ analyse.py experiments/2026-10-kappa-rule-space/scripts/w3b_ sample.py experiments/2026-10-kappa-rule-space/scripts/w3c_ analyse.py	W3c: "What sets kappa? A rule-space and mechanism programme", W3c registration block in README.md, author timestamp 2026-10-15 20:55 (+13:00), before sampling or simulation; result log timestamp 21:19 (+13:00), recorded commit da88e61. Runner: scripts/run_w3c.sh. Rule sampling seed 20261014+23; patched starts 501–508 (w3c_sample.py). The crossed training/test comparison is exploratory, not preregistered. The other seed excerpts below concern earlier stages. Seed declarations: rules do not straddle folds. Seeds: `numpy.random.default_rng(20261014)` for folds, written at `99de7c1`; other commits landed in between, none touching `data/`), seed 20261014: **Fresh sample.** 150 rules, 75 each of length n = 13 and n = 14 (numpy `default_rng(20261014 + 11)`), Registration/context records (combined result notes are not independent preregistration

Audit	Exact records and producers	Seeds, registration and Git record
	experiments/2026-10-kappa-rule-space/scripts/w3c_cross.py experiments/2026-10-kappa-rule-space/scripts/w3c_sample.py experiments/2026-10-kappa-rule-space/scripts/w4_specs.py experiments/2026-10-kappa-rule-space/scripts/w4_stats.py experiments/2026-10-kappa-rule-space/scripts/w5_analyse.py experiments/2026-10-kappa-rule-space/scripts/w5_specs.py experiments/2026-10-kappa-rule-space/scripts/w6_analyse.py experiments/2026-10-kappa-rule-space/scripts/w6_checks.py experiments/2026-10-kappa-rule-space/scripts/w6_sample.py experiments/2026-10-kappa-rule-space/scripts/w7_emergence.py experiments/2026-10-kappa-rule-space/scripts/x1_prefix_icc.py experiments/2026-10-kappa-rule-space/scripts/x2_prefix_heldout.py experiments/2026-10-kappa-rule-space/scripts/x3_w3b_checks.py	evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 35b42c749d9958419fef689d0fdfb7a52e437445 (2026-10-15T19:38:36+13:00)
13	notes/excursion-tomography-2026-10-13.md	Seed declarations: - Patched starts: seeds 0..7 (`RULE@0` .. `RULE@7`), $N = 44 \times 8 = 352$ runs. - Rotor walker: `rotor:SEED`, seeds 301..304. - Random walk: `rw:SEED`, seeds 301..304. Registration/context records (combined result notes are not independent preregistration evidence): Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 99de7c1dacd81e2e065c7acfc46d6a19ae51370a (2026-10-14T00:16:50+13:00)
14	experiments/2026-10-14-t1-excursion-tomography/ experiments/2026-10-14-t1-excursion-tomography/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-t1-excursion-tomography/scripts/t1_audit.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_depth_analysis.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_excursions.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_pair_disp.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_specs.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1b_analysis.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1b_cutoff.py	Seed declarations: Analysis seed 20261014 (no random operations); simulation seed IDs are read - `scripts/t1_specs.py`: Patched starts seeds 0..7 # Patched starts seeds 0..7 Registration/context records (combined result notes are not independent preregistration evidence): Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
15	experiments/2026-10-15-independent-area/results/analysis.json experiments/2026-10-15-independent-area/README.md Analysis scripts (commands in recipe): experiments/2026-10-15-independent-area/scripts/analyse.py experiments/2026-10-15-independent-area/scripts/make_notebook.py	Seed declarations: Bootstrap rules (2000, seed 20261015), keeping roles and windows together; controls (44 rules, 3 groups, 200 independent datasets, seed 20261015): SEED = 20261015 Registration/context records (combined result notes are not independent preregistration evidence): Open questions: hypotheses before measuring — Written 2026-10-10 10:30 NZDT. Data in hand: probe_summary.csv (c, kappa, ...

Audit	Exact records and producers	Seeds, registration and Git record
		notes/open-questions-hypotheses.md Earliest path commit: uncommitted / no tracked history
16	experiments/2026-10-14-kappa-matched-area/results/noise_floor.txt experiments/2026-10-14-kappa-matched-area/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-kappa-matched-area/scripts/matched_area.py experiments/2026-10-14-kappa-matched-area/scripts/matched_pairs.py experiments/2026-10-14-kappa-matched-area/scripts/noise_floor.py	Seed declarations: ln kappa on ln A, with rule-bootstrap 95% intervals (seed 20261014, 500 resamples). (tol 0.05 registered, 0.10 sensitivity), rule means over seeds as before. Bootstrap seed 20261016. """M1-M3 of ../README.md on the committed T1b windows. No simulation; bootstrap seed 20261014.""" Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: 741ccf3c63ec5053f31de6fb4c8ab15267158beb (2026-10-15T19:38:37+13:00)
17	notes/transport-forecast-2026-10-13.md	Seed declarations: rules 2000 times, seed 20261013; report within-rule seed sd. Full trajectory records, input SHA256 hashes, exclusions, commit and seed: 61.33–80.58% (2,000 resamples, seed 20261013). The registered 30% prediction Registration/context records (combined result notes are not independent preregistration evidence): Prospective transport forecast — Registered 2026-10-13 23:35 NZDT, before computing forecasts. Base commit notes/transport-forecast-2026-10-13.md Earliest path commit: 89d8dd6742cbf5663fdb63e5e275078c82c2d384 (2026-10-15T19:38:37+13:00)
18	experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py experiments/2026-10-13-transport-forecast/results/transport_forecast.json experiments/2026-10-13-transport-forecast/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py	Seed declarations: rng = np.random.default_rng(20261013) command='uv run experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py', bootstrap_seed=20261013, Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 797704a2988ea8d958df39ac96e3ece27bec0980 (2026-10-14T19:31:38+13:00)
19	notes/frontier-2026-10-13-results.md	Seed declarations: Statistic: x_k = A(2^k)/2^(2k/3), k = 24..30. Per k: CV with bootstrap SE (2000 resamples, seed 20261013). Estimator: local slope beta(k) = dlog P / dlog A between octaves k-4 and k (`edge_exponent.py` definition, P = boundary cells), rotor seeds 1..8 to 2^34. - Local area exponents a = log2(A(2^24)/A(2^22))/2 are 0.84-1.02 (median ~0.93; one run, RRRLR seed 2, is at 1.02), matching the earlier 0.92 for this ensemble. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Calibration against the rotor (Eulerian) walker notes/calibration.md Earliest path commit: 5be7ac7f1b29e49bba217ccc4f7010e503e86c10 (2026-10-13T21:51:43+13:00)

Audit	Exact records and producers	Seeds, registration and Git record
20	notes/walker-comparisons-2026-10-13.md	Seed declarations: Run each plus random walk and rotor controls for seeds 301..308, to 2^20, fits, input hashes, commit and seed: `experiments/2026-10-13-w-width-radius-refit/results/width_radius_refit.json`. Seed 301 selected before inspecting paths; 65,536 steps, ant RLR blank start. Registration/context records (combined result notes are not independent preregistration evidence): Walker comparisons: growth and boundary width — ## Registered 2026-10-13 23:27 NZDT, before the refit notes/walker-comparisons-2026-10-13.md Earliest path commit: 89d8dd6742cbf5663fdb63e5e275078c82c2d384 (2026-10-15T19:38:37+13:00)
21	experiments/2026-10-13-memory-walkers/results/memory_walkers_pilot/summary.json experiments/2026-10-13-memory-walkers/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-memory-walkers/scripts/memory_walkers_pilot.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 797704a2988ea8d958df39ac96e3ece27bec0980 (2026-10-14T19:31:38+13:00)
22	examples/layers.rs Analysis scripts (commands in recipe): examples/layers.rs	Seed declarations: run(r, Rotor { seed: seed.parse().unwrap() }, size, max_log2) Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: 563812d720a65e4baa3b01de5d65f9883000cbfa (2026-10-13T23:05:08+13:00)
23	notes/kappa-experiment-design.md	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: 563812d720a65e4baa3b01de5d65f9883000cbfa (2026-10-13T23:05:08+13:00)
24	experiments/2026-10-13-layers-colours/results/layers_stats.txt experiments/2026-10-13-layers-colours/results/layers_stats_typical.txt experiments/2026-10-13-layers-colours/results/layers_classes.txt experiments/2026-10-13-layers-colours/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-layers-colours/scripts/layers_census_c.py experiments/2026-10-13-layers-colours/scripts/layers_classes.py experiments/2026-10-13-layers-colours/scripts/layers_select.py experiments/2026-10-13-layers-colours/scripts/layers_stats.py experiments/2026-10-13-layers-colours/scripts/layers_stats_typical.py	Boundary-layer ant runs: deterministic blank boards, no random seed. Control seeds and full rule list: experiments/2026-10-13-layers-colours/results/layers_specs.txt. Registration: "What sets kappa? Experiment design", written 2026-10-13 before runs; exploratory crust classification was not predicted. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)

Audit	Exact records and producers	Seeds, registration and Git record
25	experiments/2026-10-13-arm-kappa-depletion-arms/results/armb_stats.txt experiments/2026-10-13-arm-kappa-depletion-arms/results/armc_stats.txt experiments/2026-10-13-arm-kappa-depletion-arms/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armb_stats.py experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armc_pairs.py experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armc_stats.py	Noisy-rotor seeds: 1, 2, 3, 4, 5, 6, 7, 8. Exact input: experiments/2026-10-13-arm-kappa-depletion-arms/results/armb_specs.txt. Producer: src/bin/probe.rs; analysis: scripts/armb_stats.py. Registration: "What sets kappa? Experiment design", Arm B (2026-10-13). Actual command and deviations are in that note; the planned board and window differed from those run. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
26	experiments/2026-10-13-mixing/results/mix_rules.txt experiments/2026-10-13-mixing/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-mixing/scripts/mixing_calib.py experiments/2026-10-13-mixing/scripts/mixing_stats.py	Seed declarations: rng = np.random.default_rng(20261012) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
27	notes/mixing-random-sample.md	Seed declarations: Protocol: draw 40 rules uniformly at random (numpy default_rng seed 20261012) Registration/context records (combined result notes are not independent preregistration evidence): Mixing test on a random sample of rules — Written 2026-10-12, before running. Reason: the earlier mixing sample notes/mixing-random-sample.md Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
28	experiments/2026-10-13-item4-exponent-rotor-calibrated/scripts/item4_stats.py experiments/2026-10-13-item4-exponent-rotor-calibrated/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-item4-exponent-rotor-calibrated/scripts/item4_select.py experiments/2026-10-13-item4-exponent-rotor-calibrated/scripts/item4_stats.py	Seed declarations: rule=pl.col("rule").str.split(SEP).list.get(0), seed=pl.col("rule").str.split(SEP).list.get(1).cast(pl.Int64)) and over the census width exponent (low/mid/high), numpy seed 20261012. Writes experiments/2026-10-13-item4-exponent-rotor-calibrated/results/item4_rules.txt and experiments/2026-10-13-item4-exponent-rotor-calibrated/results/item4_walkers.txt (RULE@SEED, seeds 1..24).""" Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
29	experiments/2026-10-13-shape-limit/scripts/shape_stats.py experiments/2026-10-13-shape-limit/results/shape_stats.txt experiments/2026-10-13-shape-limit/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-shape-limit/scripts/shape_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
30	experiments/2026-10-13-item21-slow-outliers/scripts/item21_stats.py experiments/2026-10-13-item21-slow-outliers/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-item21-slow-outliers/scripts/	Seed declarations: rule_=pl.col("rule").str.split("@").list.get(0), seed=pl.col("rule").str.split("@").list.get(1).cast(pl.Int64)) controls (0.30 < alpha_w < 0.37), numpy seed 20261013; write experiments/2026-10-13-

Audit	Exact records and producers	Seeds, registration and Git record
	item21_select.py experiments/2026-10-13-item21-slow-outliers/scripts/ item21_stats.py	item21-slow-outliers/results/item21_walkers.txt (RULE@SEED, seeds 1..12)."""" core = s.filter((pl.col("alpha_w") > 0.30) & (pl.col("alpha_w") < 0.37)).sample(30, seed=20261013) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
31	notes/frontier-results.md	Seed declarations: (seeds 1..24, `RULE@SEED`), 2^30 steps, octave probe output - H12 (formation). Patched-start ants (5x5 random patch, seeds 1..48), 2^26 from 12 random 5x5 patches (seeds 1..12), 2^27 steps, same fit window as the census Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 50206967c655891659ac 98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
32	experiments/2026-10-13-w3/results/w3_stats.txt experiments/2026-10-13-w3/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-w3/scripts/w3_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
33	experiments/2026-10-13-w-width-radius-refit/results/w_ checks.txt experiments/2026-10-13-w-width-radius-refit/results/ width_radius_refit.json experiments/2026-10-13-w-width-radius-refit/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-w-width-radius-refit/scripts/w_ checks.py experiments/2026-10-13-w-width-radius-refit/scripts/width_ radius_refit.py	Seed declarations: SEED = 20261013 Registration/context records (combined result notes are not independent preregistration evidence): Walker comparisons: growth and boundary width — ## Registered 2026-10-13 23:27 NZDT, before the refit notes/walker-comparisons-2026-10-13.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
34	experiments/2026-10-13-a4-boundary-width/results/a4c_ stats.txt experiments/2026-10-13-a4-boundary-width/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-a4-boundary-width/scripts/a4_ stats.py experiments/2026-10-13-a4-boundary-width/scripts/a4c_ stats.py	Seed declarations: rng = np.random.default_ rng(20261013) Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research- frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
35	experiments/2026-10-14-hull-width/results/hull_stats.txt experiments/2026-10-14-hull-width/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-hull-width/scripts/hull_bands.py experiments/2026-10-14-hull-width/scripts/hull_stats.py	Seed declarations: patched-start seeds 221-224 (24 runs, `RULE@SEED`), and rotor seeds 221-228 (8 runs), 8192^2 (per run); medians over runs, with between-run sd and a rule-level bootstrap (seed 20261014) for `uv run experiments/2026-10-14-hull-width/ scripts/hull_bands.py` -> `results/hull_bands.txt` (no simulation; 6 ant rules x 4 seeds, 8 rotor runs; bootstrap seed 20261014). Registration/context records (combined result notes are not independent preregistration evidence): Open questions: hypotheses before measuring — Written 2026-10-10 10:30 NZDT. Data in hand: probe_summary.csv (c, kappa, ...

Audit	Exact records and producers	Seeds, registration and Git record
		notes/open-questions-hypotheses.md Earliest path commit: 741ccf3c63ec5053f31d e6fb4c8ab15267158beb (2026-10-15T19:38:37+13:00)
36	experiments/2026-10-13-edge-statistics/scripts/edge_ exponent.py experiments/2026-10-13-edge-statistics/results/edge_ exponent.txt experiments/2026-10-13-edge-statistics/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-edge-statistics/scripts/edge_ exponent.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
37	experiments/2026-10-13-b5-late-highways/results/b5_ stats.txt experiments/2026-10-13-b5-late-highways/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-b5-late-highways/scripts/b5_ stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
38	examples/ribbon_trace.rs Analysis scripts (commands in recipe): examples/ribbon_trace.rs	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 734b6822654033f16b11 7ca68d9c46b63564f09a (2026-10-13T23:41:31+13:00)
39	experiments/2026-10-13-g1-ribbon-breakout/results/g1_ stats.txt experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_ builder_*.py experiments/2026-10-13-g1-ribbon-breakout/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-g1-ribbon-breakout/scripts/ compare_survivors.py experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_ breakout_stats.py experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_ stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
40	examples/g1_ribbon_breakout_builder_dump.rs experiments/2026-10-13-g1-ribbon-breakout/scripts/ g1_wake_extract.py	Recovered unchanged dynamics from worktree-g1-builder-structure, commit c82030e. Blank board, grid 32768, RLRLLL; T0=12800000000 and 12900000000, 400000 steps each. Reproduction: experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_builder_run.sh. Compact extracts and raw input hashes: results/ g1_wake_extract.json. Original registration recovered as Audit 41.
41	notes/g1-builder-structure-2026-10-14.md	Recovered from worktree-g1-builder-structure, commit c82030e. Contains predictions recorded before the original measurement and appended results, including a falsified step-scale prediction. One deterministic blank-board trajectory, two dump windows. Model comparison is post hoc; no third-point or independent-seed validation.
42	experiments/2026-10-15-chirality-baselines/results/ comparison.json experiments/2026-10-15-chirality-baselines/results/ groups.csv experiments/2026-10-15-chirality-baselines/README.md	uv run experiments/2026-10-15-chirality-baselines/scripts/compare.py uv run research-outputs/cube-root-growth-paper/chirality_baseline_numbers.py Critique-driven reanalysis; same 300 calibration and 2747 held-out rules. Paired rule bootstrap:

Audit	Exact records and producers	Seeds, registration and Git record
		4000 draws, seed 20261046; fits fixed. Input hashes and base commit in comparison.json; per-rule predictions and an executed notebook accompany the results.
43	experiments/2026-10-15-kappa-tail-budget/results/confirm-v1.analysis.json experiments/2026-10-15-kappa-tail-budget/results/confirm-v1.decomposition.json experiments/2026-10-15-kappa-tail-budget/README.md	uv run experiments/2026-10-15-kappa-tail-budget/scripts/run.py --name confirm-v1 uv run experiments/2026-10-15-kappa-tail-budget/scripts/confirm_analysis.py uv run experiments/2026-10-15-kappa-tail-budget/scripts/confirm_decomposition.py uv run research-outputs/cube-root-growth-paper/tail_budget_numbers.py Seeds 2026101801-32 (predictor odd, outcome even offsets); bootstrap seed 2026101599, 2000 draws. Manifest sha256 19a86779... in confirm-v1.manifest.json. Registered in the README before launch; analysis and decomposition scripts written after the run (deviation entries in NOTEBOOK.md). Dumps are gitignored; a fresh clone reruns about 94 minutes.
44	experiments/2026-10-16-kappa-uniform-baseline/README.md; results/analysis.txt; results/census_spread.txt; scripts/analyse.py; examples/kappa_uniform_baseline.rs.	Uniform-resampling control: seeds 2026101601–08, grid 4096, 2^{23} steps. Its CSV labels octave starts, k=12–22 (ends 13–23). Census uses k=26 occupation and k=20–26 exponent fits. Registration and averaging correction retained. Evidence available at revision 0de4d38.
45	experiments/2026-10-19-kappa-chiral-edge/README.md; scripts/c_law.py; results/c_law.json. Run with uv from repo root.	Census stationary blobs n=2185; 176 fresh stationary rules from the leadrun-fresh experiment, seeds 601–608, probe 27 8192. Bootstrap and split seed 20261019, 2000 resamples. Stage 3 registered before scoring fresh c. Frozen coefficients, capped lead 8; R^2(c) and R^2(ln c) are distinct scores. Evidence pinned at 0de4d38.
46	experiments/2026-10-19-c-law-mechanism/README.md; NOTEBOOK.md; results/PROVENANCE.md; stage2_analysis.json, stage2c_analysis.json, stage3_analysis.json, stage4_analysis.json under results/. Commands: scripts/run_stage2.sh, run_band.sh, run_ablation.sh, run_stage3.sh, run_p1.sh, and the stage analysis scripts (uv).	Evidence pinned at 0de4d38; original code in b7b4758 and 8cc447f. Validation sampling seed 20261020; bootstrap/permutation seed 20261019, 2000 draws; patched starts 601–608. Band exhaustion, shorter random-walk calibration and density-support deviations are disclosed. Some written registration times conflict with file times: PROVENANCE.md supplies recovered clock times; no stronger timestamp guarantee is inferred here. Stage 4c full replay requires regenerating ignored probe dumps; rerunning stage4_analyse.py without them omits 4c. Paper extraction reads saved results without overwriting them.
47	experiments/2026-10-16-kappa-square-loop/README.md; results/analysis.json; scripts/analyse_loops.py. Also the kappa-loop-surgery and kappa-growth-clock experiment READMEs of 2026-10-16.	Square-loop analysis at f26b6ef: 300 fresh rules, eight seeds 601–608, 176 stationary; probe k=26 occupation window. Registered prevalence and interior-loop mediation failed; other loop shapes/transients untested. Surgery closed unrun because its gate failed. Growth-clock support gate retained 7/16 families versus 12 required, so primary effects were not fitted. All records available at 0de4d38.
48	experiments/2026-10-19-kappa-walker-reference/README.md; results/analysis.txt, analysis_nrotor.txt, probe_mem.csv, probe_rotor.csv, probe_nrotor.csv; scripts/run.sh, run_nrotor.sh, analyse.py, analyse_nrotor.py.	Eight seeds 2026101901–08 for plain walkers, 2026102001–08 for noisy rotor. Memoryless probe 23 16384; rotor/noisy probe 27 8192. Fits k=12–22 and rotor k=20–26; census occupation is k=26 alone. Noise scan: nine levels, post-escape window deviation and drift reported. Local-exponent reference agrees within 0.006 in the plain-walker comparison. Notebook close-out at 18b79d7; evidence pinned at 0de4d38.